Pure Year 2 • Chapter 12

Vectors in three dimensions

Coordinates, magnitude and direction

Geometric reasoning

Applications to mechanics

Chapter Overview 2

1. Distance between two points

How far apart are $(1, 0, 4)$ and $(-3, 5, 9)$?

2. Column vectors and unit vectors

$$\begin{pmatrix} 1 \\ -2 \\ 5 \end{pmatrix} = \mathbf{i} - 2\mathbf{j} + 5\mathbf{k}$$

3. Magnitude and direction

Find the angles that $2\mathbf{i} - 3\mathbf{j} - \mathbf{k}$ makes with the positive coordinate axes.

4. Geometric problems

Extend the Year 1 vector methods to three dimensions.

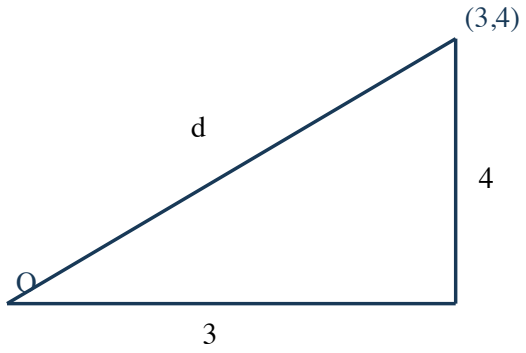
5. Applications to mechanics

Use $\mathbf{F} = m\mathbf{a}$ with three-dimensional vectors. Relate displacement to distance and velocity to speed.

This chapter reviews three-dimensional coordinates and extends familiar vector results from 2D to 3D.

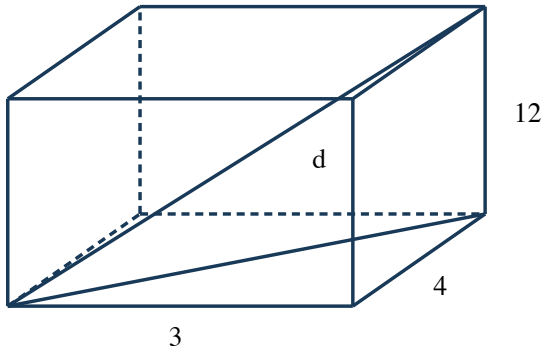
Distance from the origin and magnitude of a vector 3

In 2D, find the distance of $(3, 4)$ from the origin.



2D distance

In 3D, find the distance of $(3, 4, 12)$ from the origin.

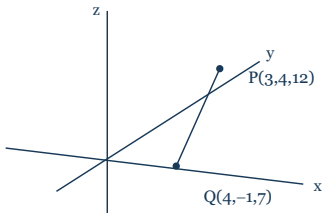


3D distance

For $\mathbf{a} = (x, y, z)^T$, its magnitude is $|\mathbf{a}| = \sqrt{x^2 + y^2 + z^2}$

Distance between two 3D points 4

For $P(3, 4, 12)$ and $Q(4, -1, 7)$, find PQ.



PQ

$d = \sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2}$. Δx means change in x. Squaring means a positive or negative coordinate change gives the same result.

Quick checks

1. Distance from $(4, 0, -2)$ to the origin.

1

2. Magnitude of $(5, 4, -1)^T$.

2

3. Distance between $(0, 4, 3)$ and $(5, 2, 3)$.

3

4. Distance between $(1, 1, 1)$ and $(2, 1, 0)$.

4

5. Distance between $(-5, 2, 0)$ and $(-2, -3, -3)$.

5

Test Your Understanding So Far... 5

1. Find the distance from the origin to $P(7, 7, 7)$.

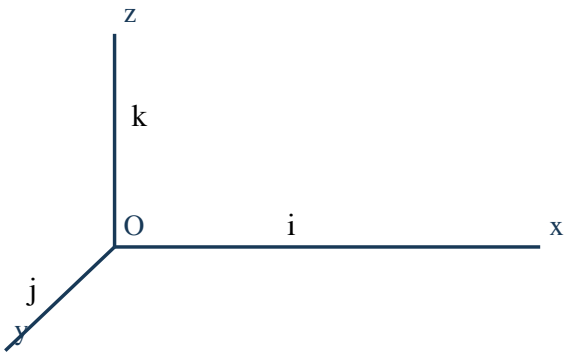
1

2. $A = (5, 3, -8)$ and $B = (1, k, -3)$. Given $AB = 3\sqrt{10}$, find the possible values of k.

2

Unit vectors in 3D 6

Unit vectors point along the positive coordinate axes. Each has magnitude 1.



In 2D:

$$\mathbf{i} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{j} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

In 3D:

$$\mathbf{i} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{j} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{k} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Every three-dimensional vector can be written as

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ y \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ z \end{pmatrix}$$

$$= x \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + y \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + z \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$= x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$$

Examples 7

Recall the 2D unit vectors:

$\mathbf{i} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{j} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$

$8\begin{pmatrix} 1 \\ 0 \end{pmatrix} - 2\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 8 \\ -2 \end{pmatrix}$

$= 8\mathbf{i} - 2\mathbf{j}.$

1. Write in $\mathbf{i}, \mathbf{j}, \mathbf{k}$ notation.

a) $(1, 2, 3)^T$

1a

b) $(-7, 3, 0)^T$

1b

c) $(3, 0, -1)^T$

1c

2. Write as a column vector.

a) $4\mathbf{j} + \mathbf{k}$

2a

b) $\mathbf{i} - \mathbf{j}$

2b

3. If $A = (1, 2, 3)$, $B = (4, 0, -1)$, find $\overrightarrow{AB}$.

3

4. If $\mathbf{a} = (2, 3, 4)^T$, $\mathbf{b} = (0, -1, 3)^T$, find $3\mathbf{a} + 2\mathbf{b}$.

4

Examples 8

1. Find the magnitude of $\mathbf{a} = 2\mathbf{i} - \mathbf{j} + 4\mathbf{k}$, then the unit vector $\hat{\mathbf{a}}$ in its direction.

1

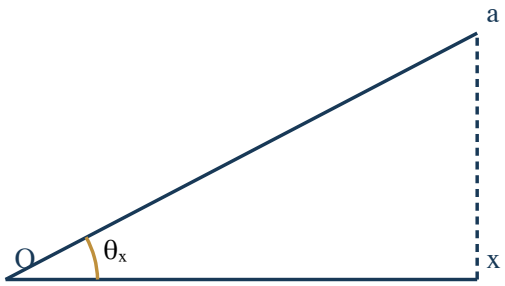
Dividing by the magnitude gives a vector of unit length without changing its direction.

2. If $\mathbf{a} = (2, -3, 5)^T$ and $\mathbf{b} = (4, -2, 0)^T$, is $2\mathbf{a} - 3\mathbf{b}$ parallel to $4\mathbf{i} - 5\mathbf{k}$?

2

Angles between vectors and an axis 9

For a non-zero vector $\mathbf{a} = (x, y, z)^T$, use a right-angled triangle to find the angle with a positive coordinate axis.



$$\cos\theta_x = \frac{x}{|\mathbf{a}|}, \quad \cos\theta_y = \frac{y}{|\mathbf{a}|}$$
$$\cos\theta_z = \frac{z}{|\mathbf{a}|}$$

Find the angles that $\mathbf{a} = 2\mathbf{i} - 3\mathbf{j} - \mathbf{k}$ makes with the positive coordinate axes.

Example

Test Your Understanding 10

1. A and B have position vectors $4\mathbf{i} + 2\mathbf{j} + 7\mathbf{k}$ and $3\mathbf{i} + 4\mathbf{j} - \mathbf{k}$. Find $\overrightarrow{AB}$ and show that triangle OAB is isosceles.

1

2. Find the angle between $\mathbf{a} = 2\mathbf{i} + \mathbf{j} + \mathbf{k}$ and the positive x-axis.

2

3. Repeat for $\mathbf{b} = \mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$.

3

4. Why is the angle between $\mathbf{a}$ and $\mathbf{b}$ not the difference of these angles?

4

Angle diagram

13. The position vectors of A, B, C are $(8, -7, 4)^T$, $(8, -3, 3)^T$, $(12, -6, 3)^T$.

a) Find $\overrightarrow{AB}, \overrightarrow{AC}, \overrightarrow{BC}$.

a

b) Find the exact lengths of all three sides.

b

c) Describe triangle ABC.

c

14. $A = (3, 4, 8)$, $B = (1, -2, 5)$, $C = (7, -5, 7)$.

a) Find $\overrightarrow{AB}, \overrightarrow{AC}, \overrightarrow{BC}$.

a

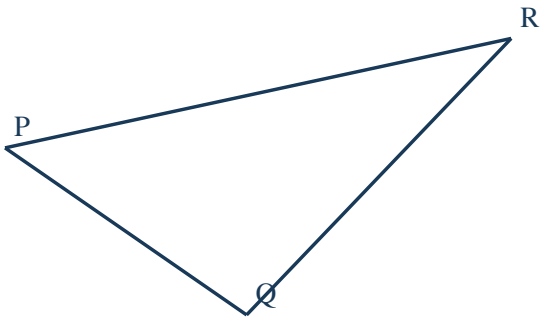
b) Find the side lengths of triangle ABC.

b

c) Given $\angle ABC = 90°$, find $\angle BAC$.

c

17. In triangle PQR, $\overrightarrow{PQ} = 3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ and $\overrightarrow{QR} = -2\mathbf{i} + 4\mathbf{j} + 3\mathbf{k}$. Show that $\angle PQR = 78.5^\circ$ to 1 d.p.



Side lengths

Cosine rule

1. $A = (2, -5, -8)$, $B = (1, -7, -3)$, $C = (0, 15, -10)$, $D = (2, 19, -20)$.

a) Find $\overrightarrow{AB}$ and $\overrightarrow{DC}$ in unit-vector notation.

1a

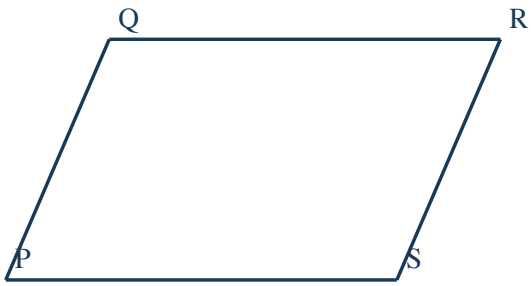
b) Show that $\overrightarrow{DC} = 2\overrightarrow{AB}$, and hence the lines are parallel.

1b

c) Describe quadrilateral $ABCD$.

1c

2. $P = (4, -9, -3)$, $Q = (7, -7, -7)$, $R = (8, -2, 0)$. Find S so that $PQRS$ is a parallelogram.



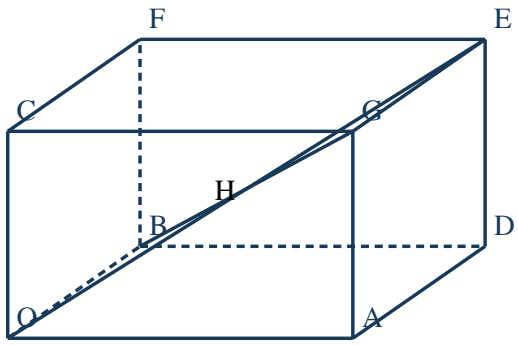
2

Compare coefficients just as for a polynomial identity: $3x^2 + 5x \equiv A(x^2 + 1) + Bx + C$ gives $A = 3$ from the x^2 terms.

1. Find p, q, r if
$3\mathbf{i} + (p + 2)\mathbf{j} + 120\mathbf{k}$
$= p\mathbf{i} - q\mathbf{j} + 4pqr\mathbf{k}$

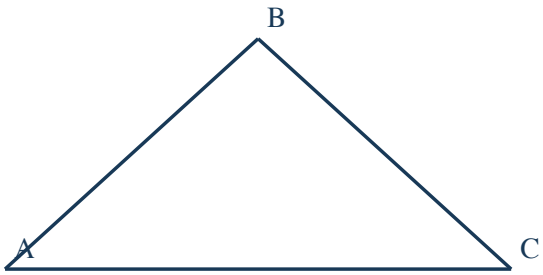
1

2. A cuboid has $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$, $\overrightarrow{OC} = \mathbf{c}$. Prove diagonals OE and BG bisect each other.



2

6. Triangle ABC has $\overrightarrow{AB} = 7\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ and $\overrightarrow{BC} = -\mathbf{i} + 5\mathbf{k}$.



a) Find its area. (7 marks)

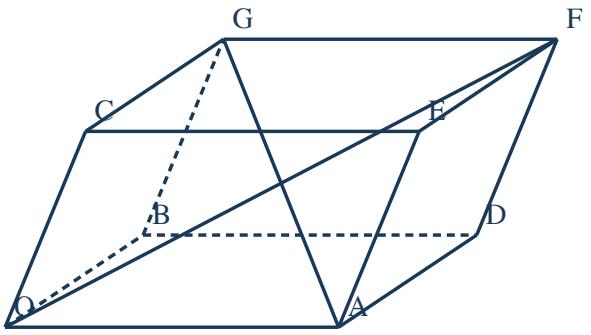
a

Points D, E satisfy $\overrightarrow{AD} = 3\overrightarrow{AB}$ and $\overrightarrow{AE} = 3\overrightarrow{AC}$.

b) Find the area of triangle ADE. (2 marks)

b

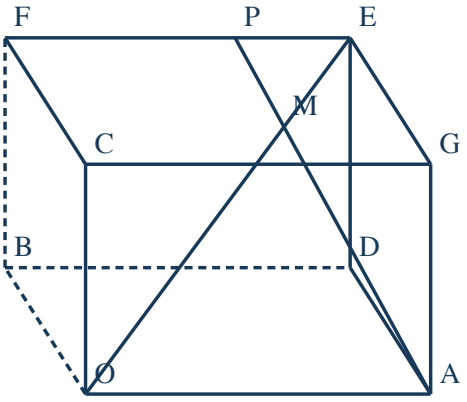
7. A parallelepiped is a solid with six parallelogram faces. The diagram has vertices O, A, B, C, D, E, F, G, with $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$, $\overrightarrow{OC} = \mathbf{c}$. Prove that OF and AG bisect each other.



Two routes

Conclusion

8. In the cuboid, $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$, $\overrightarrow{OC} = \mathbf{c}$. Point M lies on OE, with $OM : ME = 3 : 1$. Line AP passes through M, with $AM : MP = 3 : 1$. Prove that P lies on EF and find $FP : PE$.



Locate P

Position and ratio

Application to Mechanics 19

Displacement, velocity, acceleration and force are vectors. Speed, mass and time are scalars. Find the magnitudes below.

1. Force: $(3, 4, -1)^T$ N.

1

2. Acceleration: $(1, 0, 1)^T$ m s^{-2}.

2

3. Displacement: $(12, 3, 4)^T$ m.

3

4. Velocity: $(0, 4, 3)^T$ m s^{-1}.

4

The magnitude of displacement is the straight-line separation. It equals distance travelled only for motion along that line without reversing.

Example 20

A particle of mass 0.5 kg is acted on by three forces:

$$\mathbf{F}_1 = (2\mathbf{i} - \mathbf{j} + 2\mathbf{k})\ \text{N}$$
$$\mathbf{F}_2 = (-\mathbf{i} + 3\mathbf{j} - 3\mathbf{k})\ \text{N}$$
$$\mathbf{F}_3 = (4\mathbf{i} - 3\mathbf{j} - 2\mathbf{k})\ \text{N}$$

a) Find the resultant force.

b) Find the acceleration in unit-vector notation.

c) Find its magnitude.

d) Starting from rest, find the distance travelled in the first 6 seconds.

a

b

c

d

2. A particle starts from rest and accelerates at $(4\mathbf{i} - 2\mathbf{j} + 3\mathbf{k})\,\mathrm{m\,s^{-2}}$ for 2 seconds. Find the distance travelled.

2

4. A particle of mass 6 kg is acted on by $\mathbf{F}_1$ and $\mathbf{F}_2$. Given $\mathbf{F}_1 = (7\mathbf{i} + 3\mathbf{j} + \mathbf{k})\,\mathrm{N}$ and $\mathbf{a} = (2\mathbf{i} - \mathbf{k})\,\mathrm{m\,s^{-2}}$, find $\mathbf{F}_2$.

4

8. The diagram shows parallelogram $ABCD$. Given $\overrightarrow{AB} = (6, -2, 11)^T$ and $\overrightarrow{AC} = (15, 8, 5)^T$, find its area. (7 marks)



Sides

Area